

JACK GILBERT J

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EDUCATION

Bachelor of Technology in Computer Science Engineering 2021 – 2025 | Coimbatore
Karunya Institute Of Technology and Science

MAJOR UNIVERSITY PROJECTS

Real-Time Coconut Tree Disease Detection Using YOLOv8 and Explainable AI May 2025

- Developed a real-time disease detection system using YOLOv8, capable of identifying multiple coconut tree diseases (e.g., bud rot, gray leaf spot) from image and video inputs with high accuracy.
- Integrated Explainable AI (EigenCAM) to generate visual heatmaps, providing farmers with transparent insights into AI predictions for better trust and decision-making.
- Built a Streamlit-based user interface for ease of use by non-technical users, supporting image uploads, webcam feeds, and RTSP streams for remote monitoring.
- Achieved effective performance on low-resource devices; included automated disease management recommendations, contributing to reduced crop loss and enhanced agricultural productivity.

Lane Lines Detection using Python and OpenCV Mar 2023

- Developed an image and video-based lane detection system using OpenCV and Python to identify road lane boundaries.
- Implemented preprocessing steps including color filtering, grayscale conversion, Gaussian blur, and Canny edge detection.
- Applied region of interest masking and Hough Line Transform to detect and visualize lane lines accurately.
- Averaged and extrapolated lane lines for robustness across diverse road scenarios.
- Demonstrated effectiveness on static images and video clips, contributing to autonomous driving and driver assistance technologies.

CERTIFICATIONS

Introduction to Data Science, Cisco Networking Academy  Sep 2023

- Completed 30 hours lectures and tutorials covering foundational concepts in data science, including data wrangling, visualization, and statistical analysis.
- Gained hands-on experience with Python, Jupyter Notebooks, and real-world datasets, building a solid base for machine learning and data-driven decision-making.

Master the Fundamentals and Advanced Concepts, Scaler  Oct 2023

- Completed course for ~11.5 months covering Data Structures & Algorithms, System Design, SQL, and Low-Level Design.
- Gained hands-on experience with Java, Spring Boot, React, and Redux, culminating in a capstone project integrating backend and frontend technologies.

Google Cybersecurity, Coursera  Apr 2024

- Completed 8 courses of ~170 hours covering essential cybersecurity concepts, including threat detection, network security, SIEM tools, and risk management.
- Gained hands-on experience with tools like Wireshark, Linux, SQL, and Splunk, and conducted real-world security analyses as part of the program's final capstone project.

Google Data Analytics, Coursera  Apr 2024

- Completed 8 courses for 180 hrs focusing on data cleaning, analysis, and visualization using tools like SQL, Excel, R, and Tableau.
- Applied data analytics process to real-world case studies, including data preparation, hypothesis testing, and dashboard creation, culminating in a capstone project.

Course on Ethical Hacking, Cisco Networking Academy  Oct 2024

- Acquired hands-on skills in ethical hacking methodologies, reconnaissance, network scanning, and system exploitation using industry-relevant tools.
- Successfully completed tutorials covering penetration testing, vulnerability assessment, and common attack vectors.

Meta Android Developer, Coursera  Oct 2024

- Completed 8 self-paced courses (approximately 286 hours) covering Android development fundamentals, including Kotlin programming, Android Studio, UI/UX design, and mobile architecture.
- Developed and deployed a capstone project—a food ordering app—demonstrating proficiency in MVVM architecture, LiveData, and API integration.

Course on Intro to Machine Learning, Kaggle  Oct 2023

Course on Data Visualization, Kaggle  Sep 2023

SKILLS

Computer Languages — Python, Matlab, SQL, C, Linux

Data Science — Machine Learning, Deep Learning, Data Engineering, Data Analytics, Statistics

Data Visualization — Excel, Tableau, Power BI, Google Analytics Studio, Matplotlib, Seaborn

Soft Skill — Solution Centric Thinker, Creative, Resilient, Promptitude, Self Monitoring, Adaptive

INTERESTS

Data Analytics | Machine Learning | Deep Learning | Data Engineering | Data Visualization | Data Management |
Business Intelligence | Market Research | Data Scavenging